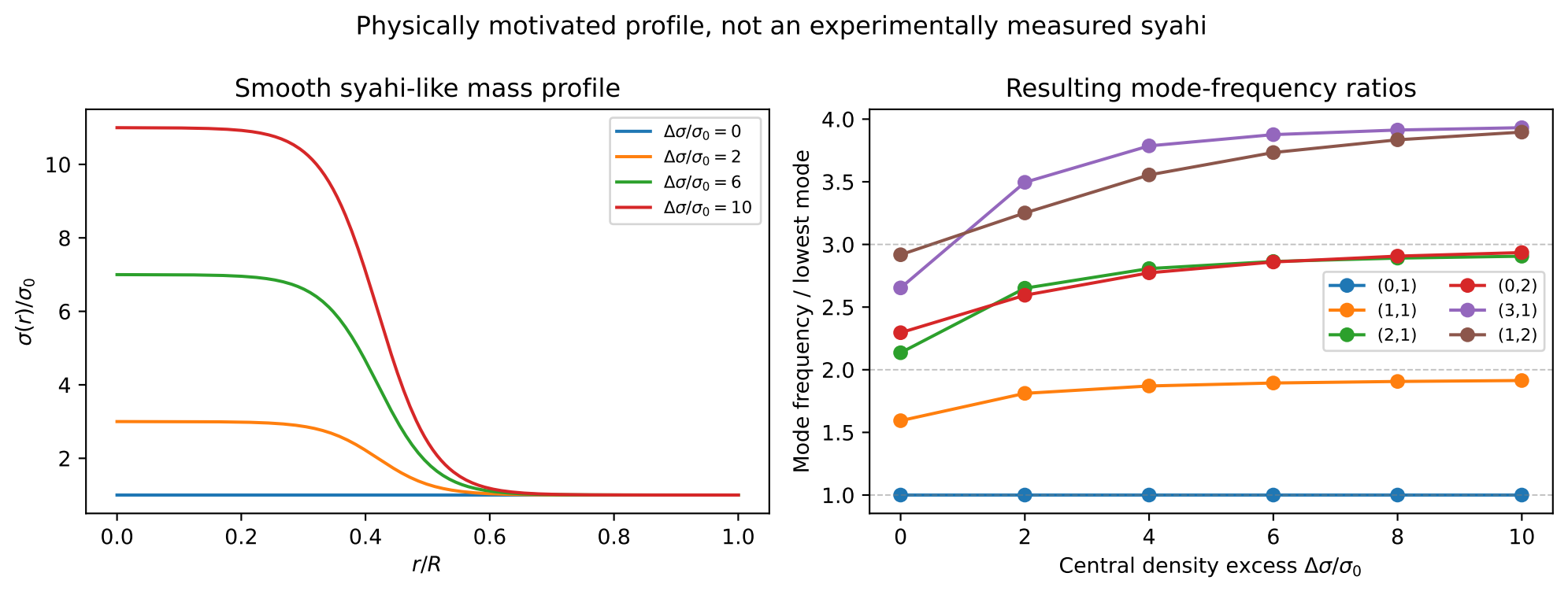
Physically motivated profile, not an experimentally measured syahi
Smooth syahi-like mass profile
$\sigma(r)/\sigma_0$
r/R
$\Delta\sigma/\sigma_0 = 0$
$\Delta\sigma/\sigma_0 = 2$
$\Delta\sigma/\sigma_0 = 6$
$\Delta\sigma/\sigma_0 = 10$
Resulting mode-frequency ratios
Mode frequency / lowest mode
Central density excess $\Delta\sigma/\sigma_0$
(0,1)
(1,1)
(2,1)
(0,2)
(3,1)
(1,2)